Generative Δ-Seekers for Circuit Design Discovery

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***Abstract*—Generative models are increasingly being used to automate circuit synthesis and optimization by proposing topologies, sizing devices, and exploring large design spaces. We show how a generative model can instead be placed inside a closed-loop search, allowing AI-driven design to move from automation toward *design discovery*. We propose Δ-Seeker, where Δ denotes a verified departure from known designs in circuit structure and electrical behavior rather than structural novelty alone. Δ-Seeker combines a discrete diffusion model for topology generation and mutation with structural checking, SPICE-in-the-loop sizing, and Pareto selection over performance and structural novelty. A state-dependent controller uses search progress and recurring structural failures to decide where additional evaluation effort should be spent and when later diffusion generations should be steered away from failure patterns observed earlier. Independent re-sizing and operating-corner analysis are performed only after the search. In generator-only experiments with no sizing or Pareto selection, cross-generation feedback increases structurally valid children from 143 to 945 while reducing the faulty-candidate fraction from 0.925 to 0.761, showing that earlier search results change what the generator produces. State-dependent steering limits the mean increase in faulty candidates after intervention to $+0.009$, compared with $+0.096$ under periodic steering. In amplifier discovery at comparable sizing cost, the closed-loop search returns 99 topologies, of which 54 remain feasible in at least three of four independent re-sizings and 18 also pass physical screening and supply–temperature corner analysis, compared with 38 and 10 among 59 topologies returned by fixed-control search. Removing cross-generation memory reduces independently robust discoveries from 54 to 43 and supply–temperature robust discoveries from 40 to 27. In DC–DC converter search, nearest-parent analysis identifies a corpus-absent topology whose generated freewheel path changes its nearest training-corpus parent from infeasible at all four operating corners to feasible at all four, while increasing median efficiency from 31.2% to 98.96%. These results show how generative circuit models can move beyond automated design generation toward closed-loop discovery of circuit structures that are both distinct from known designs and electrically meaningful.**

***Index Terms*—Topology synthesis, design-space exploration, diffusion models, analog circuits, DC–DC converters**

I. Introduction

Generative models are increasingly being used to automate circuit synthesis and optimization by learning from existing designs and using that knowledge to propose topologies, size devices, and explore large design spaces. Graph neural networks [1], language models [2]–[5], reinforcement learning [6], and diffusion models [7], [8] have all been applied to analog and power-converter design. In this work, **we investigate how generative design can move from *design automation* toward *design discovery***, where the goal is not only to generate feasible circuits, but to find structures that depart from known designs while retaining useful, and potentially improved, electrical behavior. This is difficult because moving beyond the known design space rapidly increases the risk of invalid connectivity, failed simulation, and poor performance.

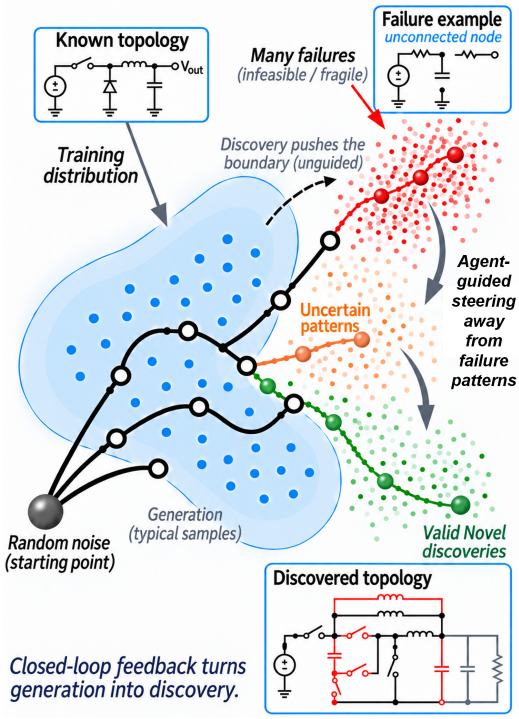


Fig. 1. From generation to discovery. Generative sampling returns typical designs from the learned distribution, while discovery pushes beyond it, where novelty introduces greater risk of invalid or fragile circuits. Closed-loop feedback steers later generations away from recurring failures toward valid novel designs.

We introduce Δ-Seeker, a closed-loop generative circuit discovery method in which Δ denotes a verified departure from known designs in both circuit structure and operating behavior. A diffusion model generates and mutates topologies, while structural checks, SPICE-in-the-loop sizing, and Pareto selection evaluate feasibility, performance, and structural novelty. A state-dependent controller uses search progress and recurring failures to allocate additional evaluation effort and steer later generations. Failures therefore affect not only which candidates survive, but also what is generated next. Unlike LLM-based EDA methods that directly generate or correct circuits [9], [10], select and size designs in multi-agent flows [11]–[13], reuse design knowledge [14], or operate design tools [15], [16], the controller does not generate or evaluate circuits itself.

We study Δ-Seeker on two circuit discovery problems, op-amp and DC-DC converter topology discovery, which present different discovery challenges. In op-amps, departures from known structures can quickly lead to invalid or poorly per-

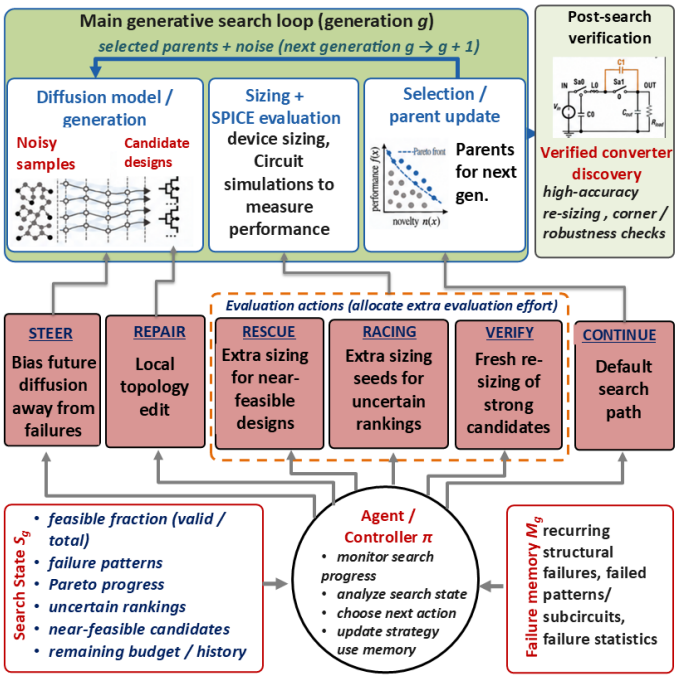


Fig. 2. Overview of Δ-Seeker. A generative search loop combines diffusion-based topology generation, sizing and SPICE evaluation, and Pareto selection. A state-dependent controller uses search state and failure memory to select evaluation, repair, or steering actions, followed by independent post-search verification.

forming circuits, while in converters, structurally distinct graphs may be inactive, redundant, or functionally equivalent to known designs. In op-amp topology discovery, where departures from known designs can quickly lead to invalid or poorly performing circuits, search feedback increases the number of topologies that survive independent re-sizing from 38 to 54 and the number that also pass device-level checks and supply-temperature corners from 10 to 18. The discovered designs include one topology combining a composite input branch, flipped-voltage-follower bias, and push-pull load drive with 27.2–29.0 dB gain, about 7 dB above a common-source reference, and another that drives the PMOS load gate from the source-degeneration node and loses 2.3 dB when that generated connection is held fixed. In DC–DC converter discovery, nearest-parent analysis identifies three corner-robust topologies absent from the training corpus. Two reduce to a known buck converter with parallel inductors, while the third introduces a capacitor that creates a new freewheel path, changing its nearest parent from infeasible at all four operating corners to feasible at all four and increasing efficiency from 31.2% to 98.96%. Our contributions are:

- Δ-Seeker, a closed-loop generative circuit discovery method that uses search feedback to adapt both circuit evaluation and subsequent topology generation
- a cross-generation feedback mechanism that uses recurring failures and search progress to guide later exploration, with controlled experiments showing changes in the generated distribution and improved discovery yield
- circuit-level validation across op-amp and DC–DC converter discovery using independent re-sizing, operating-corner analysis, device-level checks, circuit ablations, nearest-parent comparison, and waveform analysis to separate structural novelty from verified changes in circuit operation.

II. Closed-Loop Generative Circuit Discovery

Δ-Seeker operates as an iterative generate–evaluate–adapt loop. A diffusion model proposes circuit topologies, which are checked for structural validity, sized, and simulated in SPICE. The search records high-performing designs, candidates close to feasibility, and structural patterns that repeatedly fail. This information is used to allocate additional sizing or repair effort to current candidates and to steer subsequent generations away from recurring failures. The loop therefore adapts both search effort and topology generation as exploration proceeds. After optimization, selected designs are independently re-sized and verified to determine whether the generated structural changes produce reproducible circuit behavior. The key components of Δ-Seeker are described below.

A. Topology Generation and Mutation. We represent each circuit as a component–net graph in which component terminals connect through net nodes and the resulting topology is converted directly into a SPICE netlist. For amplifiers, circuits are assembled from reusable motifs extracted from the training corpus, while DC–DC converters are represented directly using switches, inductors, and capacitors. A circuit motif is a recurring subcircuit, such as a bias branch or transistor arrangement, that serves as a reusable building block during generation. A discrete diffusion model p_θ is trained on valid circuit graphs to learn their structural distribution without using simulated circuit performance.

The first generation is obtained by denoising fully corrupted graphs. In later generations, selected circuits serve as parents and nearby structural variants are generated by partially corrupting and denoising them again

$$K_\gamma(x' \mid x) = \int q(x_{\gamma T} \mid x) \prod_{t=\gamma T}^{1} p_\theta(x_{t-1} \mid x_t, c)\, dx_{1:\gamma T}, \quad (1)$$

where q is the forward corruption process, T is the diffusion length, and γ controls mutation strength. Small γ produces local topology changes, while γ approaching one moves generation toward a new sample from the learned distribution. Structural constraints during denoising discourage invalid terminal connections and dangling nets. This forms the basic generate–mutate process. Δ-Seeker adds feedback across generations so that failure patterns observed in earlier candidates can alter the topologies proposed later.

B. Candidate Evaluation and Search. Each generated topology is first checked for structural validity and SPICE feasibility. Valid candidates are then sized using an inner NSGA-II loop [17]. The performance objective $f(x)$ is amplifier gain for op-amps and conversion efficiency for DC–DC converters, while a second objective measures structural novelty relative to the training set. The formulation readily extends to

additional design objectives and constraints. For converters, we use Weisfeiler–Lehman features $\phi(x)$ and define

$$n(x) = 1 - \max_{y \in \mathcal{D}} \cos\big(\phi(x), \phi(y)\big), \tag{2}$$

where $\mathcal{D}$ contains the structurally distinct training circuits. The amplifier search follows the same graph-similarity principle. The outer loop uses Pareto selection over circuit performance and structural novelty. Selected designs are mutated with (1) to form the next generation, while topologies reaching the Pareto front are retained in the search archive.

C. Search Feedback and Adaptive Control. After each generation, Δ-Seeker summarizes the current search status, including feasible candidates, Pareto progress, near-feasible designs, uncertain rankings, and recurring structural failures. It also records previous search actions and the remaining evaluation budget. We denote these by the search state S_g, action history H_g, failure memory M_g, and remaining budget B_g. The next operation is selected as

$$a_g = \pi\left(S_g, H_g, M_g, B_g\right). \tag{3}$$

The selected operation is executed through the same structural-checking, sizing, and SPICE flow, and its result updates the search state before the next generation. The policy π is implemented with an LLM that selects from a fixed set of search operations. The LLM neither generates netlists nor evaluates circuit performance.

For existing candidates, RESCUE allocates additional sizing effort to designs near feasibility, RACING evaluates additional sizing seeds when rankings are uncertain, VERIFY re-sizes strong candidates from fresh initializations, and REPAIR applies a bounded local topology modification followed by structural checking, sizing, and simulation. CONTINUE follows the default search path. STEER instead adapts the generation process for the next generation. Failure patterns accumulated in M_g bias later diffusion steps away from recurring faulty structures, with strength λ_g; $\lambda_g = 0$ recovers the unbiased sampler. STEER may raise or lower λ_g according to the observed search state and may also adjust the mutation strength γ in (1) and the number of children generated per parent. Changes to γ and offspring count apply to the next generation, whereas λ_g remains active until the next STEER. The diffusion model itself is not retrained, so failures observed in generation g can alter the topologies proposed in generation $g + 1$.

D. Verification. Search-time metrics alone are not used to establish discovery. After optimization, selected topologies are re-sized from fresh initial conditions and re-simulated independently of the search loop. These verification results are not used to update the controller. For op-amps, we check repeated sizing convergence, device-body and bias connectivity, signal-current paths, gate-bias conditions, and operation across supply and temperature corners with device sizes fixed.

For DC–DC converters, selected topologies are evaluated across four operating corners and compared with their nearest circuits in the training set. A topology is classified as corpus-absent only when no equivalent graph is found after canonicalization, including checks after removal of one or two devices. Parent–child simulation, switching-waveform analysis, and equivalent-circuit reduction are then used to determine whether the generated structural modification changes circuit operation. These post-search checks establish the Δ in Δ-Seeker as a structural departure from known designs that also produces a reproducible change in circuit behavior.

III. OP-AMP TOPOLOGY DISCOVERY

We first evaluate Δ-Seeker on op-amp topology discovery. The search targets transistor-level amplifier structures assembled from reusable circuit motifs, with the objective of finding topologies that provide useful gain while remaining feasible after device sizing and SPICE simulation and that differ structurally from circuits in the training set. The key challenge is that moving farther from known amplifier structures rapidly increases the probability of invalid connectivity, failed operating points, or poor circuit performance. We therefore evaluate whether search feedback can increase the number of structurally distinct amplifier topologies that survive fresh re-sizing and supply–temperature corner analysis, rather than simply increasing the number of search candidates.

A. Experimental Setup. All configurations use the same PDK, diffusion checkpoint, structural checks, `ngspice` evaluator, sizing flow, feasibility criteria, Pareto selection, and post-search verification. Every structurally valid topology receives an initial sizing run, while candidates that produce a usable gain response can receive deeper sizing.We compare methods at approximately 1,000 sizing evaluations per seed rather than at a fixed generation count because adaptive search can allocate additional sizing and re-evaluation according to the observed search state. Base diffusion samples directly from the learned prior, fixed control uses predetermined search rules,periodic..Periodic steering uses the same operation set but applies STEER every five generations not adaptive .. Δ-Seeker without memory retains adaptive evaluation but removes cross-generation failure memory, and full Δ-Seeker combines adaptive evaluation with memory-guided generation. All configurations use three seeds. Full Δ-Seeker runs for 20 generations and submits 3,024 candidates for simulation.

The seven op-amp verification conditions are nominal, $V_{\mathrm{DD}} \pm 10\%$, $-40\,^{\circ}\mathrm{C}$, $125\,^{\circ}\mathrm{C}$, $V_{\mathrm{DD}} - 10\%$ at $125\,^{\circ}\mathrm{C}$, and $V_{\mathrm{DD}} + 10\%$ at $-40\,^{\circ}\mathrm{C}$. Device sizes are fixed across corners. We report both bias re-trimming at each corner and the stricter case in which the nominal bias is also frozen.

B. Discovery Yield. Under the common evaluation flow in Table I, full Δ-Seeker submits 3,024 candidates, of which 2,872 (95%) pass structural checking and 1,629 produce usable gain after sizing. The search places 99 distinct topologies in the Pareto archive. Independent re-sizing confirms 54 designs, compared with 38 under fixed control, and 40 remain feasible at all seven supply–temperature conditions with device sizes frozen and the bias re-trimmed. Eighteen designs pass both the device and bias checks and all seven corners, compared with 10 under fixed control. The independent re-sizing pass rates are 55% for Δ-Seeker and 64% for fixed control, so

TABLE I

OP-AMP SEARCH UNDER A COMMON PDK, SPICE, SIZING, AND VERIFICATION FLOW. FRESH RE-SIZE DENOTES DESIGNS FEASIBLE IN AT LEAST 3 OF 4 INDEPENDENT SIZING RUNS. V, T DENOTES FEASIBILITY AT ALL SEVEN CORNERS WITH BIAS RE-TRIMMING. FINAL DENOTES DESIGNS THAT ALSO PASS DEVICE AND BIAS CHECKS.

Method	Sizing eval.	Valid (%)	Pareto archive	Fresh re-size	V, T	Final
Base diffusion	3,027	22	14	8	5	3
Fixed control	3,074	92	59	38	31	10
Δ-Seeker, no memory	3,008	92	81	43	27	12
Δ-Seeker	3,395	95	99	54	40	18

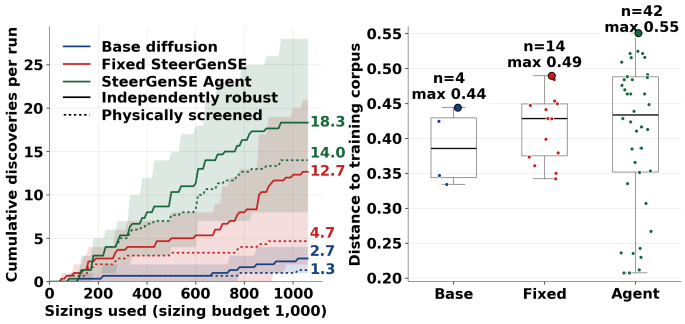


Fig. 3. Discovery yield under a fixed 1,000-sizing budget. **Left**, cumulative independently re-sized and device-checked designs as sizing evaluations are consumed. **Right**, structural distance of the device-checked designs from the training corpus.

the higher verified count comes from reaching a larger set of reproducible designs rather than from a looser post-search filter. Median structural novelty is 0.626 for Δ-Seeker and 0.617 for fixed control. The median difference between search-time gain and independently re-sized gain is also small, $+0.94$ dB for Δ-Seeker and $+1.46$ dB for fixed control.

A separate experiment fixes the budget to exactly 1,000 sizing evaluations per run and measures discovery as search effort is consumed. Δ-Seeker reaches 18.3 independently re-sized and 14.0 device-checked designs per run, compared with 12.7 and 4.7 under fixed control and 2.7 and 1.3 under base diffusion. The median structural distance of the device-checked designs is 0.43 for both Δ-Seeker and fixed control and 0.39 for base diffusion, indicating that the higher yield is not obtained by moving closer to the training set.

Across the ten verified designs in Table II, gain spans 27.4–37.9 dB, GBW spans 2.6–885.3 MHz, and phase margin spans 53°–96°, showing that the search does not collapse to a single amplifier structure or operating regime. All ten designs pass all seven supply–temperature conditions when the input bias is re-trimmed, while only two to four of the seven conditions pass with both device sizes and bias frozen. We therefore report the two corner criteria separately because robustness of the generated topology and device sizing does not imply robustness of a fixed bias point.

C. Representative Op-Amp Topologies and Discovered Circuit-Level Concepts. Table II summarizes ten verified op-amp topologies discovered by Δ-Seeker spanning different

TABLE II

REPRESENTATIVE OP-AMP DISCOVERIES AFTER INDEPENDENT RE-SIZING AND DEVICE-LEVEL CHECKS. FRESH REPORTS FEASIBLE RUNS OUT OF FOUR RE-SIZINGS; CORNERS REPORT FEASIBLE CONDITIONS WITH RE-TRIMMED BIAS / FROZEN SIZE AND BIAS. NOVELTY IS THE WL DISTANCE TO THE NEAREST TRAINING TOPOLOGY. ⋆ MARKS TOPOLOGIES SHOWN IN FIG. 4.

Design	Dev.	Gain dB	GBW MHz	PM °	Power mW	Fresh	Corners trim/froz.	Nov.
D1⋆	4M	32.1	120.9	92	3.36	4/4	7/7, 2/7	0.393
D2⋆	5M+4	30.5	62.2	96	9.80	3/4	7/7, 3/7	0.464
D3⋆	7M	37.9	885.3	53	31.49	3/4	7/7, 2/7	0.328
D4	4M	34.0	257.5	89	42.27	3/4	7/7, 3/7	0.228
D5⋆	5M	33.8	2.6	91	14.71	3/4	7/7, 4/7	0.303
D6	8M	31.6	481.7	88	287.20	3/4	7/7, 3/7	0.411
D7	6M+2	29.4	84.9	73	5.02	3/4	7/7, 2/7	0.429
D8	3M+1	29.3	189.2	94	11.30	4/4	7/7, 2/7	0.388
D9	4M	27.8	108.8	96	6.84	3/4	7/7, 2/7	0.365
D10	4M+1	27.4	207.8	88	14.43	3/4	7/7, 2/7	0.352

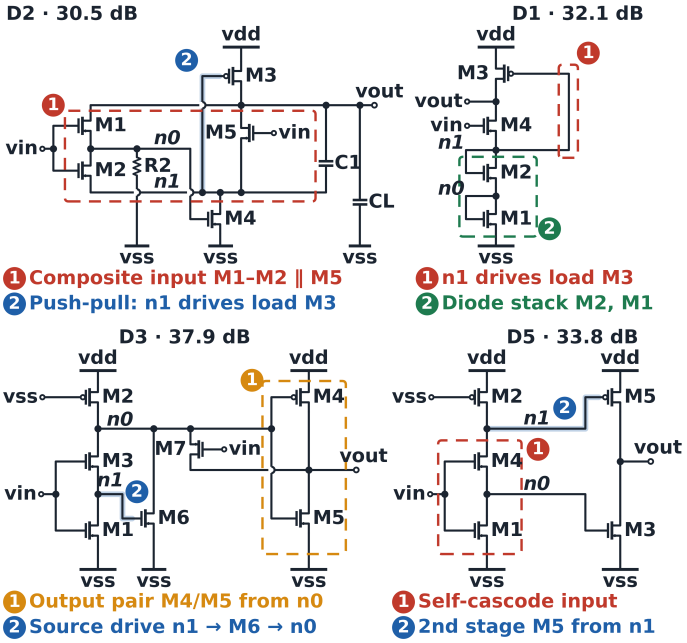


Fig. 4. Two verified Δ-Seeker op-amps and their independently re-simulated AC responses after re-sizing. The left design combines a composite input branch, flipped-voltage-follower bias, and push-pull load drive. The right design drives the PMOS load gate from the source-degeneration node.

gain, bandwidth, phase margin, power, and structural distance from the training set. Fig. 4 examines two specific generated topologies to determine what circuit mechanisms their new connections realize and whether those connections contribute to operation.

- ***T1, composite input branch with flipped-voltage-follower bias and push-pull load drive.*** The generated topology in Fig. 3(a) combines three recognizable circuit mechanisms. The M1–M2 branch forms a series-parallel composite transistor [18] in parallel with M5. Biasing M4 from the branch midpoint makes M2 and M4 operate as a flipped voltage follower [19], while the resulting node drives the PMOS load to provide push-pull operation [20]. For the three independently re-sized operating points used in the

circuit-level analysis, the topology achieves 27.2–29.0 dB gain at 0.46 mA, approximately 7 dB above the common-source reference. Replacing the M4 bias connection with an equivalent diode-connected implementation produces similar gain, supporting its role as a biasing mechanism rather than the source of the gain improvement. Thus, the result is not a new primitive circuit element, but a generated combination of established mechanisms whose structure differs from the nearest training examples and whose operation is confirmed through independent sizing and simulation.

- ***T2, push-pull load drive through the source-degeneration node.*** The second topology uses diode-connected M2 and M1 to degenerate input device M4, while the resulting source-degeneration node also drives the PMOS load gate. This generated connection couples the input-stage degeneration node directly to the load-drive function, realizing push-pull operation through a different signal path. After independent re-sizing, the topology achieves 26.0–30.4 dB gain at 0.99 mA. A connection ablation with device sizes held fixed reduces gain by 2.3 dB, directly isolating the contribution of the generated load-drive connection at the verified operating point. Push-pull load drive itself is established, but its realization through the source-degeneration node is a distinct circuit-level structure identified by the search.

D. Effect of Cross-Generation Memory. We first isolate structural memory with generator-only sampling, without sizing, Pareto selection, or search control. Cross-generation memory reduces recurrence of previously recorded failure patterns from 17.6% to 15.7%, lowers the fraction of samples with structural faults from 0.925 to 0.761, and increases cumulative structurally valid children from 143 to 945 per seed. Because no downstream selection is present, this experiment shows that failure history changes the generated topology distribution.

The full-search ablation keeps adaptive evaluation but removes only cross-generation memory. Restoring memory increases the Pareto archive from 81 to 99 designs, independently re-sized designs from 43 to 54, and seven-corner designs from 27 to 40. Device and bias checks alone give 21 designs with memory and 23 without memory, so the ablation supports improved robust discovery yield without implying improvement at every intermediate verification stage.

E. Effect of State-Dependent Control. State-dependent control and periodic steering use the same generator, failure memory, and operation set but differ in when intervention is applied. In Fig. 6, Δ-Seeker applies STEER early when feasibility is low and later shifts effort to RESCUE, VERIFY, and RACING. The faulty-candidate fraction changes by at most 0.01 after each state-dependent STEER operation in the representative run, while periodic STEER is followed by increases of 0.05–0.17.

Across the intervention study, state-dependent STEER occurs at a mean feasible fraction of 0.16 and is followed by a mean faulty-candidate change of +0.009, compared with +0.096 under periodic steering. Both settings use the same generator, failure memory, steering mechanism, and available operations, so the comparison isolates the effect of state-dependent intervention timing rather than introducing a different generation mechanism. The periodic-search totals in Table I are not used for this comparison because that configuration was completed for only two full search seeds.

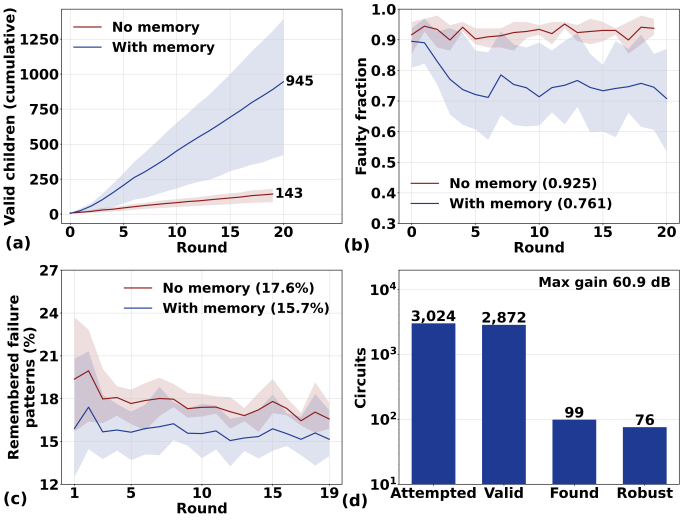


Fig. 5. Effect of cross-generation failure memory. Panels (a)–(c) show generator-only sampling over 20 rounds and three seeds. Panel (d) shows the full-search progression for Δ-Seeker; the bar labelled Robust is the end-of-run re-score criterion (≥2/4), for which 76 of 99 archive designs pass. Independent fresh re-sizing (≥3/4) confirms 54 designs in Table I.

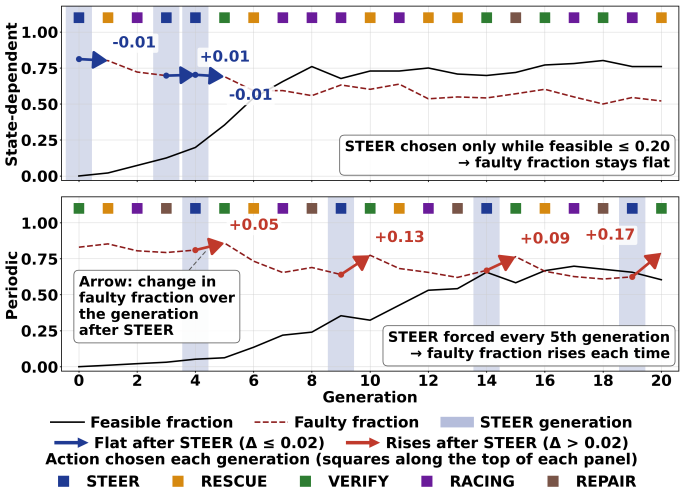


Fig. 6. Search trajectories under state-dependent control and periodic steering for the same seed. Solid and dashed curves show feasible and faulty-candidate fractions. Markers show selected operations and shaded regions indicate STEER.

F. Comparison Scope. Reported results from prior analog topology generators use different datasets, PDKs, objectives, and validity definitions [4], [5], [9]. We therefore use them as methodological context rather than as numerical baselines. The discovery-yield claims above are based on the controlled configurations in Table I, which share the same generator checkpoint, PDK, SPICE flow, sizing procedure, and verification criteria.

IV. DC–DC Converter Topology Discovery

The converter study tests whether structural novelty corresponds to a change in circuit operation. With only three to six devices per topology, graph edits can produce corpus-absent structures that are inactive, redundant, or equivalent to known converters. We therefore verify generated topologies using operating corners, nearest-parent simulation, switching-waveform analysis, and equivalent-circuit reduction.

Converter Search Setup. The LaMAGIC corpus [2] contains phase-a switches active for duty fraction D, complementary phase-b switches active for $1 - D$, inductors, and capacitors connected between V_{IN}, V_{OUT}, ground, and internal nets. After removing 1,959 records with a missing rail, 196,376 graphs with three to six devices are split into 176,738 training, 9,818 validation, and 9,820 test graphs. Repeated duty labels reduce the training set to 37,514 structurally distinct graphs by WL hash. The diffusion model is trained directly on these device–net graphs with duty as the conditioning variable. At $D = 0.5$, 71–83% of random diffusion samples pass structural checks, but none of 100 random samples meets the 2:1 conversion target and roughly 20–30% are pass-through structures. Thus, structural validity alone does not imply correct conversion.

Candidates are simulated in `ngspice` at $V_{\mathrm{IN}} = 100\,\mathrm{V}$ and 100 kHz with $L = 500\,\mu\mathrm{H}$ and $0.25\,\Omega$ series resistance, $C = 10\,\mu\mathrm{F}$ with $30\,\mathrm{m}\Omega$ ESR, $50\,\mathrm{m}\Omega$ switch on-resistance, and $R_{\mathrm{load}} = 50\,\Omega$. Feasibility requires V_{OUT} within 5% of 50 V with a valid duty-cycle response, ripple no greater than 5%, and efficiency of at least 90%. Duty is determined by a five-point sweep followed by up to two secant refinements. Verification uses nominal, heavy-load, light-load, and low-line conditions, with L and C reduced by 20% at low line.

Structural checks reject unbound pins, floating nets, shoot-through switch pairs, shorted outputs, and missing input-to-output paths. The search uses population 48 for 20 generations with efficiency and structural novelty as objectives. After search, four-corner designs are canonicalized with fixed rail assignments and compared with the training graphs, including checks after removal of one or two devices. Corpus-absent designs are then simulated together with their nearest training-corpus parents. Current probes, load and line sweeps, component-tolerance checks, and equivalent-circuit reduction are used to determine the function of the generated elements.

Verified Converter Designs. Across two seeds, Δ-Seeker finds 15 designs that satisfy all four operating corners. Three remain absent from the training corpus after graph canonicalization and are shown in Fig. 7 and Table III. All three are buck-family variants, consistent with the available device set and 2:1 step-down target.

- ***Design (a), generated freewheel path.*** The generated capacitor C_1 is active over 97% of the switching period, with $I_{\mathrm{rms}}/I_{\mathrm{load}} = 1.47$ and 6.5 A peak current at the nominal condition. Waveform analysis shows that C_1 carries the L_0 current during the interval in which the nearest training-corpus parent lacks a freewheel path. The generated connection therefore changes the parent from a non-functioning topology into a four-corner converter. Its median efficiency remains slightly below the textbook buck reference.
- ***Designs (b) and (c), active but reducible modifications.*** The generated elements in both designs are active, with utilization of at least 0.48 at the nominal condition. In design (b), they establish a conduction path for L_2 in both switching phases and remove the large voltage excursions present in the parent. Design (c) similarly repairs a parent that produces no output. Equivalent-circuit reduction, however, maps both generated topologies to a known buck with three parallel inductors. Their improved operation therefore results from repairing defective parent structures rather than introducing a distinct converter function.

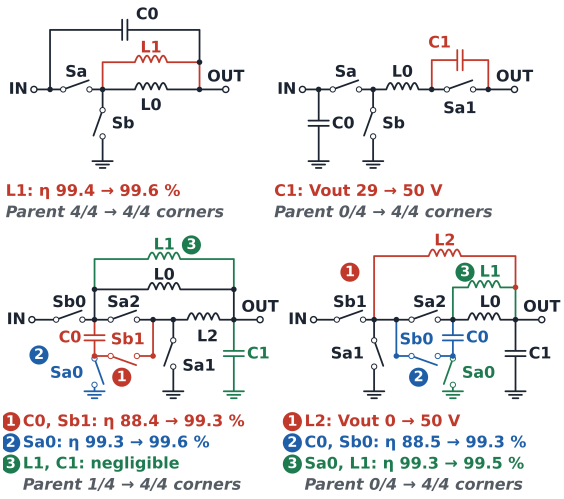


Fig. 7. Three corpus-absent converter topologies satisfying all four operating corners. Red elements are added relative to the nearest training-corpus parent and grey denotes the load.

TABLE III
PARENT–CHILD COMPARISON FOR THE FOUR CONVERTERS IN FIG. ?? (D1–D4: LEFT TO RIGHT, TOP TO BOTTOM) ACROSS FOUR OPERATING CORNERS.

Design	Generated change	Corners	Med. η (%)	Worst η (%)	Ripple (mV)
D1	L_1 parallel to L_0	$4/4 \to 4/4$	$99.37 \to 99.60$	$98.80 \to 99.28$	$44 \to 88$
D2	C_1 across S_{a1}	$0/4 \to 4/4$	$31.18 \to 98.96$	$16.82 \to 97.99$	$251 \to 88$
D3	C_0, S_{b1} link; S_{a0}, L_1, C_1	$1/4 \to 4/4$	$88.43 \to 99.64$	$79.76 \to 99.41$	$147 \to 131$
D4	L_2; C_0, S_{b0} link; S_{a0}, L_1	$0/4 \to 4/4$	$0 \to 99.45$	$0 \to 99.08$	$-- \to 131$
Ref.	Textbook buck	4/4	99.37	98.80	87

Graph Novelty versus Circuit Operation. The results separate three different outcomes that graph novelty alone cannot distinguish. Some generated graphs differ from corpus circuits only through inactive structures and do not change operation. Designs (b) and (c) contain active generated elements but reduce to a known circuit form. Design (a) instead changes the operating path of its nearest training-corpus parent, with C_1 carrying inductor current during the interval in which the parent lacks a freewheel path. A capacitor in parallel with a switch is

itself established in zero-voltage quasi-resonant converters [21], so we do not claim a new converter family. The result is a corpus-absent topology with a verified change in operation relative to its nearest training example. This distinction between structural difference and verified circuit behavior is the Δ tested by the post-search analysis.

V. Conclusion

Δ-Seeker uses search results to adapt both evaluation effort and subsequent topology generation. In op-amp discovery, this increases the number of designs surviving independent resizing and corner verification, while circuit-level analysis identifies generated combinations and connections whose operation can be tested directly. Generator-only and intervention studies show separately that cross-generation memory changes the sampled topology distribution and that state-dependent steering improves when that feedback is applied. In converter discovery, nearest-parent simulation and equivalent-circuit reduction distinguish inactive graph changes, active but functionally equivalent modifications, and a corpus-absent topology with a verified change in its freewheel path. These results show how closed-loop search and post-search circuit analysis can move generative EDA from producing candidate structures toward identifying reproducible departures from known designs.

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